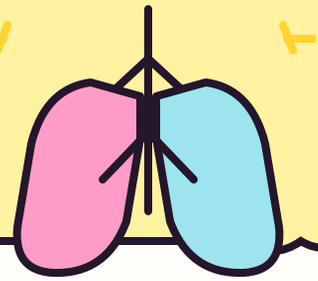


CHAPTER 19 • MED-SURG I

Chapter 19 Respiratory Disorders

Management of Patients with Chest and Lower Respiratory Tract Disorders



What This Chapter Covers

- Atelectasis and acute tracheobronchitis
- Pneumonia, aspiration, and pulmonary tuberculosis
- Lung abscess and sarcoidosis
- Pleural disease, respiratory failure, and ARDS
- Pulmonary vascular disorders, occupational disease, lung cancer, and chest trauma

Respiratory Words to Know

- Hypoxemia: low oxygen in arterial blood
- Hypercapnia: excess carbon dioxide in blood
- Dyspnea: difficult or uncomfortable breathing
- Pleuritic pain: sharp pain linked to breathing
- Hemoptysis: coughing up blood

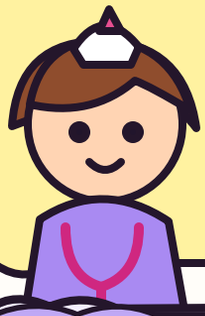
Assessment Starts with Trends

- Respiratory rate, work of breathing, lung sounds, oxygen saturation, and vital signs
- Cough pattern and secretions: amount, color, and odor
- Mental status, fatigue, intake, hydration, and activity tolerance
- Compare findings with the patient's baseline and the change over time

Escalate Deterioration

- New restlessness or confusion can signal poor oxygenation or worsening illness
- Increasing respiratory effort, tachypnea, or falling oxygen saturation needs prompt reassessment
- Cyanosis and respiratory arrest are late danger signs
- Follow orders and facility escalation procedures; reassess response to interventions

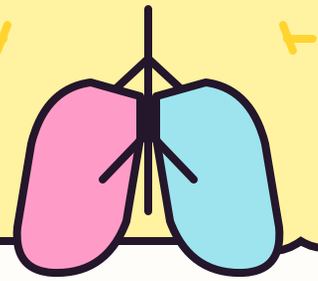
Source: Hinkle Chapter 19, slides 1-2



CHAPTER 19 • MED-SURG I

Atelectasis

Alveoli close or collapse, reducing effective ventilation



What Happens

- Closure or collapse of alveoli; may be acute or chronic
- May be obstructive or nonobstructive
- Poorly ventilated areas can impair oxygenation
- A chest X-ray may suggest atelectasis before symptoms appear

Assessment Cues

- Increased work of breathing and hypoxemia
- Decreased breath sounds and crackles over the affected area
- Pulse oximetry may show oxygen saturation below 90 percent
- Trend breathing effort and oxygenation, not one finding alone

Prevent and Expand

- Turn frequently and promote early mobilization
- Encourage voluntary deep breathing, coughing, and prescribed incentive spirometry
- Use secretion-management strategies and prescribed inhaled therapy
- A pressurized metered-dose inhaler is listed among the slide interventions
- Coordinate an ICOUGH-style multidisciplinary approach when used by the facility

If First-Line Measures Are Not

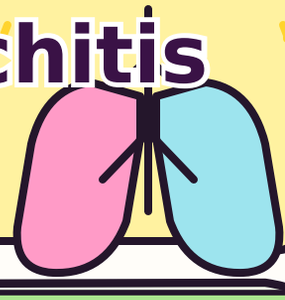
- Improve ventilation and remove secretions; first-line care includes turning, ambulation, lung expansion, and coughing
- Possible ordered therapies: PEEP or CPAP, chest physiotherapy, bronchoscopy, or mechanical ventilation
- Thoracentesis may relieve compression from pleural fluid
- Reassess oxygenation and response after each intervention

Source: Hinkle Chapter 19, slides 3-7



CHAPTER 19 • MED-SURG I

Acute Tracheobronchitis



Inflammation of the trachea and bronchi, usually after a viral infection

Pathophysiology

- Inflamed mucous membranes produce increased, often mucopurulent, sputum
- Cough may begin dry, then become productive with mucoid sputum
- As illness progresses, dyspnea, wheezes, stridor, or purulent sputum may occur

Medical Management in the Slides

- Antibiotics, when indicated by the cause and prescription
- Analgesics and increased fluid intake when safe
- Cool-vapor therapy or steam inhalation
- Suctioning when needed to manage secretions

Nursing Care

- Support bronchial hygiene and rest
- Promote fluids as appropriate to the patient's condition
- Teach the patient to complete prescribed medication courses
- Monitor for worsening breathing, oxygenation, or signs of pneumonia

Antibiotic Check

- Identify the disease process and whether antibiotics are recommended
- Do not treat a viral illness with antibiotics unless a bacterial infection is also suspected or confirmed
- Check allergies, contraindications, cultures when ordered, teaching, and nursing monitoring
- A change in sputum color alone does not identify the organism or select the drug

Source: Hinkle Chapter 19, slides 8-9; Chapter 9 antibiotic review



CHAPTER 19 • MED-SURG I

Pneumonia Types and Risks

Infection and inflammation of lung parenchyma



Causes and Categories

- **Possible causes include bacteria, mycobacteria, fungi, and viruses**
- **The slides list community-acquired, health care-associated, hospital-acquired, and ventilator-associated pneumonia**
- **The lecture notes that health care-associated pneumonia may involve multidrug-resistant organisms and needs early recognition**
- **HAP develops 48 hours or more after hospitalization; VAP follows at least 48 hours of ventilation**
- **Immunocompromised patients may have subtle onset and progressive dyspnea, fever, and a dry cough**

Community-Acquired

- **Occurs in the community or within the first 48 hours after hospitalization**
- **The slide set identifies Streptococcus pneumoniae as a common adult cause**
- **Infection rates rise with age**
- **Viral causes are common in infants and children**

Risk Factors from the Slides

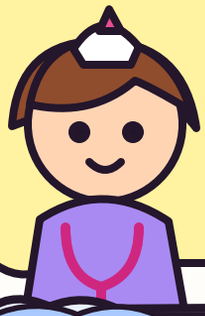
- **Heart failure, diabetes, alcoholism, COPD, AIDS, and cystic fibrosis**
- **Influenza and other respiratory infections**
- **Debility, dehydration, impaired cough, or immune suppression may increase vulnerability**
- **HAP can involve multidrug-resistant organisms; prevention and early recognition matter**

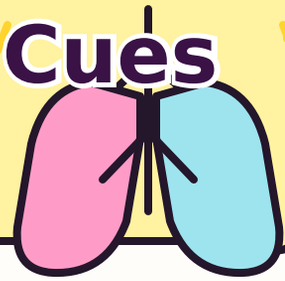
HAP, VAP, and Immune Suppression

- **HAP begins at least 48 hours after admission and has high mortality in the slide set**
- **HAP may present with fever, tachycardia, pleural effusion, or scant sputum in debilitated patients**
- **VAP prevention is emphasized; use the facility's VAP bundle**
- **Immunocompromised-host examples include Pneumocystis, fungi, and tuberculosis**

Source: Hinkle Chapter 19, slides 10-14

Pneumonia Clinical Cues





Findings vary with organism, severity, and underlying illness

Possible Manifestations

- Streptococcal pneumonia may begin suddenly with chills, fever, pleuritic pain, tachypnea, and respiratory distress
- Viral, Mycoplasma, or Legionella pneumonia may show relative bradycardia
- Other findings can include URI symptoms, headache, low-grade fever, myalgia, rash, or pharyngitis
- Orthopnea, crackles, increased tactile fremitus, and purulent sputum may occur

Assess the Whole Patient

- Trend vital signs, cough frequency and severity, and shortness of breath
- Describe secretions by amount, odor, and color
- Inspect and auscultate the chest; note oxygenation and work of breathing
- Assess mental status, fatigue, edema, hydration, intake, and coexisting heart failure

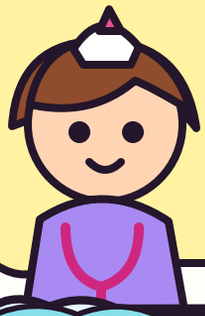
Older-Adult Watch

- New mental-status changes may accompany pneumonia
- Also monitor fatigue, dehydration, edema, and functional change
- Atypical or subtle findings do not rule out serious infection
- Compare with baseline and report concerning change promptly

Diagnostic Workup

- History and physical examination
- Chest X-ray
- Blood cultures and sputum examination when ordered
- Bronchoscopy may be used for acute severe infection

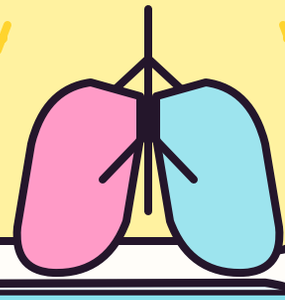
Source: Hinkle Chapter 19, slides 15-16 and 20



CHAPTER 19 • MED-SURG I

Pneumonia Treatment and Prevention

Treat the cause, support gas exchange, and watch for deterioration



Medical Management

- Use an appropriate antibiotic for bacterial pneumonia; cultures and sensitivity can guide selection
- Antibiotics do not treat viral pneumonia unless a secondary bacterial infection is present
- Supportive measures in the slides: fluids, oxygen for hypoxia, antipyretics, antitussives, decongestants, and antihistamines
- Match the plan to severity, patient factors, and the identified cause

Pneumococcal Prevention

- Lecture: routine vaccination at age 65; risk-based vaccination from age 19
- Current CDC guidance: adults age 50 and older, plus younger adults at increased risk
- Choice and timing depend on prior doses and patient-specific risk
- Use the current CDC schedule because the lecture age cutoff is older

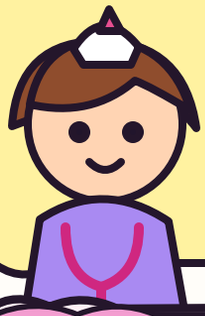
COVID-19 Notes in the Slides

- SARS-CoV-2 illness ranges from asymptomatic to severe viral pneumonia
- Symptoms include fatigue, myalgia, congestion, sore throat, diarrhea, and smell or taste loss
- Supportive outpatient care includes rest, hydration, and antipyretics
- Severe illness can raise VTE risk and progress to shock or respiratory failure

Nursing Priorities

- Position for breathing; give prescribed oxygen and reassess response
- Support coughing, pulmonary hygiene, position changes, and incentive spirometry
- Encourage hydration, nutrition, rest, activity as tolerated, and self-care teaching
- Use isolation and infection-prevention steps according to the organism and policy

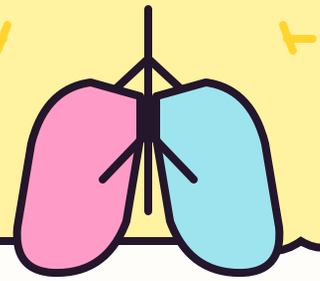
Source: Hinkle Chapter 19, slides 17-19; CDC vaccine guidance



CHAPTER 19 • MED-SURG I

Bacterial Pneumonia Complications

Recognize worsening illness and protect airway, oxygenation, and circulation



Complication Watch

- Symptoms that continue after therapy begins
- Sepsis or septic shock
- Respiratory failure, atelectasis, or pleural effusion
- Delirium, especially when accompanied by other signs of deterioration

Goals of Care

- Improve airway patency and activity
- Maintain appropriate fluid volume and nutrition
- Understand the treatment plan and prevention strategies
- Avoid complications and plan for home, community, or transitional care

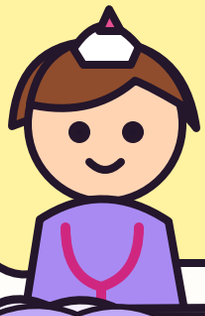
Expected Outcomes

- Improved airway patency
- Rest and energy conservation, followed by gradual activity increase
- Adequate hydration and dietary intake
- Patient describes and follows management strategies without complications

Antibiotic Safety

- Administer the prescribed drug on schedule; obtain ordered cultures first when feasible without delaying urgent treatment
- Check allergies and monitor for rash, diarrhea, or other adverse effects
- Evaluate clinical response and follow culture results or treatment changes
- Teach the patient to complete the prescribed course and report concerning reactions

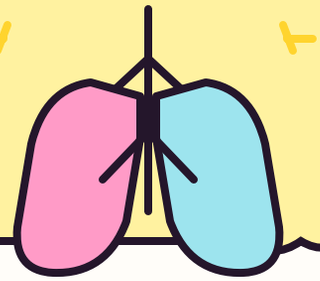
Source: Hinkle Chapter 19, slides 21-24; Chapter 19 antibiotic review



CHAPTER 19 • MED-SURG I

Aspiration

Foreign material entering the lungs can disrupt ventilation and oxygenation



Pathophysiology

- Inhaled material triggers inflammation, hypoventilation, and ventilation-perfusion mismatch
- Risk is affected by the volume and character of aspirated contents, often gastrointestinal material
- A serious complication is bronchial or lobar pneumonia
- Loss of consciousness is identified as a risk factor in the slides

Prevention Before Feeding

- Complete a swallowing screen as indicated
- Keep the head of bed elevated; maintain the endotracheal cuff if the patient is intubated
- Check feeding-tube placement before tube feeding according to policy
- Use the ordered diet and swallowing recommendations

Safer Feeding

- The slide set lists a soft diet, small bites, and no straws
- Avoid unnecessary stimulation of the gag reflex during suctioning or procedures
- Observe for cough, choking, or other swallowing difficulty
- Follow the patient's prescribed texture, positioning, and supervision plan

If Bacterial Infection Develops

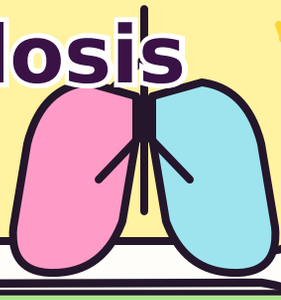
- Aspiration itself does not automatically mean bacterial pneumonia or an antibiotic indication
- The supplied exam review names ampicillin-sulbactam or amoxicillin-clavulanate as examples when anaerobic bacteria may be involved
- Drug choice depends on the clinical picture, cultures, allergies, and prescriber plan
- Monitor respiratory status and response to treatment

Source: Hinkle Chapter 19, slides 25-26; Chapter 19 antibiotic review



CHAPTER 19 • MED-SURG I

Pulmonary Tuberculosis



Airborne infection requiring testing, prevention, and sustained multidrug treatment

Organism and Spread

- Caused by Mycobacterium tuberculosis
- Spreads through the air; infection can affect the lungs and other organs
- Granulomas and a Ghon tubercle are named in the lecture
- Follow current airborne precautions and public-health direction for suspected or active TB

Common Cues

- Low-grade fever, cough, and fatigue
- Cough may be nonproductive or mucopurulent; hemoptysis can occur
- Night sweats and weight loss
- Older adults may have atypical manifestations

Assessment and Diagnosis

- History and physical examination
- Mantoux skin test: interpret by risk-based criteria, not redness alone
- TB blood testing, sputum testing, and sputum culture
- Assess for possible extrapulmonary disease and relevant exposure risks

Treatment and Nursing Care

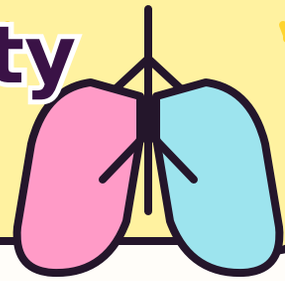
- Lecture exam cue: classic initial RIPE therapy is rifampin, isoniazid, pyrazinamide, and ethambutol
- The lecture describes an initial 8-week phase followed by a continuation phase; treatment is multidrug and adherence is essential
- Promote airway clearance, activity, nutrition, and completion of therapy; prevent transmission
- Current CDC guidance includes shorter regimens for eligible patients; the TB team chooses the regimen using susceptibility, conditions, and interactions

Source: Hinkle Chapter 19, slides 27-32; Chapter 19 antibiotic review; CDC TB guidance

TB Medication Safety



Know the classic RIPE names and what changes require prompt reporting



RIPE Memory

- **R - rifampin**
- **I - isoniazid**
- **P - pyrazinamide**
- **E - ethambutol**
- **The supplied Chapter 9 review also lists rifapentine**

Monitor for Serious Reactions

- **Liver warning signs: yellow skin or eyes, dark urine, abdominal pain, nausea, poor appetite, or unusual fatigue**
- **Isoniazid can cause peripheral neuropathy; report tingling, numbness, or pain**
- **Ethambutol can affect vision and color vision; report changes promptly**
- **Rifampin and rifapentine can interact with many medicines**

Teaching and Follow-Up

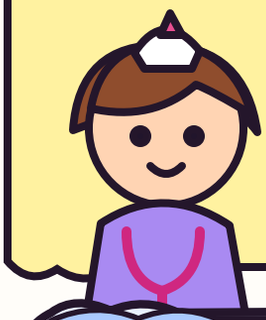
- **Do not stop or skip TB medicines without the TB care team's direction**
- **Give an accurate medication list and attend scheduled follow-up**
- **CDC recommends at least monthly clinical evaluation for adherence and adverse reactions during TB disease treatment**
- **Avoid interpreting the classic lecture regimen as the only current option; follow TB expertise and susceptibility results**

Current Practice Note

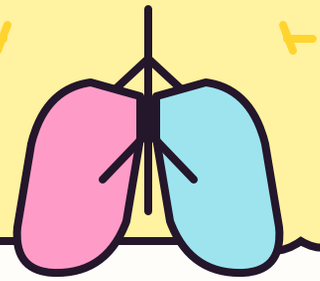
- **The source slides state 6-12 months and four or more initial drugs**
- **CDC guidance now includes a 4-month regimen for eligible adults and adolescents with drug-susceptible pulmonary TB**
- **Other patients may need a different 6- or 9-month regimen, based on clinical factors**
- **For class: know the course's classic RIPE framework; for care: use current TB-program guidance**

Source: Chapter 19 course documents; CDC adverse-event guidance

Lung Abscess



A localized infection that may follow bacterial pneumonia



Causes and Symptoms

- Most lung abscesses are complications of bacterial pneumonia
- Location relates to gravity and patient position
- Productive cough with foul sputum, pleurisy, dyspnea, weakness, anorexia, and weight loss
- The course can range from mild productive cough to acute illness

Possible Complications

- Empyema
- Bronchopleural fistula
- Pleural friction rub and crackles may be found
- Monitor clinical response and promptly report worsening respiratory status

Assessment and Diagnosis

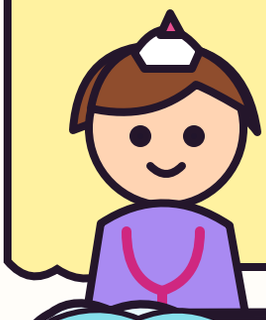
- Chest X-ray and CT of the chest
- Sputum culture
- Bronchoscopy may be used
- Use the patient's symptoms, examination, and test results together

Treatment and Nursing Care

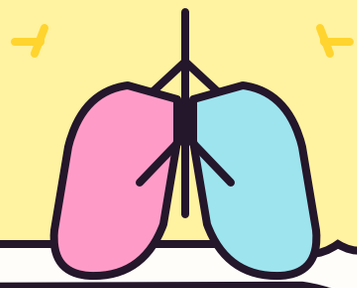
- Support drainage, chest physiotherapy, and deep breathing and coughing
- The slides list antimicrobial therapy and IV antibiotics
- The supplied review names ampicillin-sulbactam or amoxicillin-clavulanate as examples when anaerobes may be involved
- Encourage high-protein, high-calorie nutrition and provide emotional support
- Pulmonary resection is rare but appears as an option in the slides

Source: Hinkle Chapter 19, slides 33-36; Chapter 19 antibiotic review

Sarcoidosis



Inflammatory, multisystem granulomatous disease of unknown origin



Disease Snapshot

- Any organ may be affected; findings depend on the systems involved
- The lecture highlights onset often between ages 20 and 40
- The slide set notes higher occurrence among African American women
- Some cases may remit spontaneously without treatment

Possible Respiratory and Systemic

- Dyspnea, cough, hemoptysis, or congestion
- Anorexia, fatigue, and weight loss
- Symptoms vary with organ involvement
- Assess chronic-illness impact and patient concerns

Diagnostic Findings

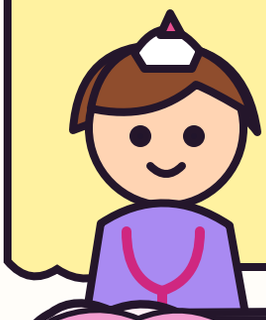
- Chest X-ray or CT scan
- Mediastinoscopy or transbronchial biopsy
- Pulmonary function testing and arterial blood gases
- Needle biopsy for definitive diagnosis in the slide set

Management and Teaching

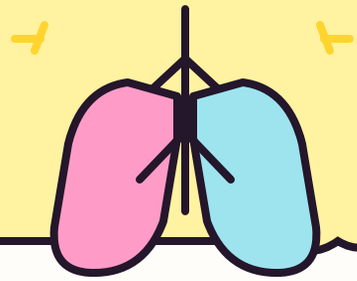
- Corticosteroids or an immune modulator may be used
- Support prescribed treatment and chronic-illness management
- Teach medication use and when to notify the primary provider
- Connect patients with community resources, including the Foundation for Sarcoidosis Research

Source: Hinkle Chapter 19, slides 37-40

Pleurisy



Inflammation of both pleural layers with pain linked to breathing



What Is Happening

- The pleura cover the lungs and line the chest wall
- Pleurisy means inflammation of both pleural layers
- Pleuritic pain changes with respiratory movement
- Pain can make deep breathing and coughing difficult

Assessment Cues

- Sharp pleuritic pain related to breathing
- A pleural friction rub may be heard with a stethoscope
- Assess breathing pattern, cough, pain, and oxygenation
- Look for the underlying condition causing pleural inflammation

Diagnostic Tests in the Slides

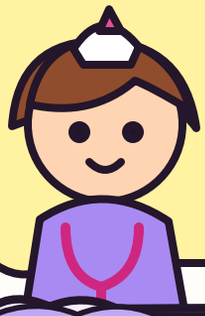
- Chest X-ray
- Sputum analysis
- Thoracentesis when indicated
- Use findings to evaluate the cause and guide treatment

Treatment and Nursing Care

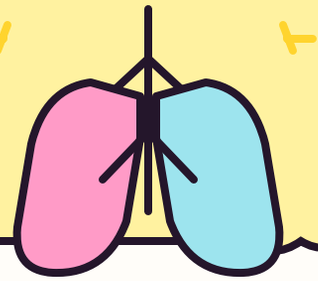
- Treat the underlying cause
- Provide prescribed analgesia
- Teach the patient to splint the rib cage when coughing
- Support comfortable lung expansion and report worsening symptoms

Source: Hinkle Chapter 19, slides 41-42

Pleural Effusion



Fluid collects in the pleural space, often secondary to another illness



Causes and Symptoms

- May occur secondary to heart failure, TB, pneumonia, or pulmonary infection
- Fever, chills, pleuritic pain, and dyspnea can occur; dyspnea is more likely with a large effusion
- The amount of fluid and underlying disease affect the presentation
- Treating the cause is part of management

Physical Assessment

- Decreased or absent breath sounds over the affected area
- Decreased tactile fremitus
- Dull or flat percussion note
- Trachea may deviate away from the affected side

Diagnosis

- Chest X-ray or CT of the chest
- Thoracentesis with fluid analysis
- Monitor breathing and oxygenation while diagnostic workup proceeds
- Report increasing dyspnea or acute respiratory distress promptly

Management and Teaching

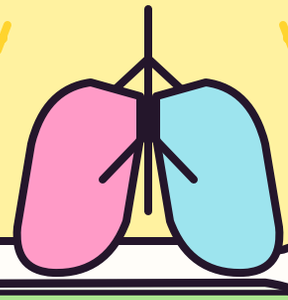
- Treat the underlying cause
- Pleurodesis may be used in selected situations
- Support the prescribed medical regimen
- Provide patient and family education and reassess respiratory status

Source: Hinkle Chapter 19, slides 41 and 43-44



Empyema

Thick purulent fluid accumulates in the pleural space



How It Develops

- May complicate bacterial pneumonia or a lung abscess
- The patient may appear acutely ill with infection-like findings
- Chest auscultation may show decreased or absent breath sounds over the affected area
- Empyema is not simply a dry pleural inflammation; purulent fluid is present

Assessment and Diagnosis

- Assess acute respiratory infection or pneumonia findings
- Chest CT
- Diagnostic thoracentesis
- Monitor respiratory status and response to drainage and medication

Treatment

- Drain the pleural fluid
- The lecture specifies antibiotics for 4-6 weeks
- The supplied exam review emphasizes culture- and source-guided antibiotic selection
- Broad IV antibiotics may be started while cultures are pending, as prescribed

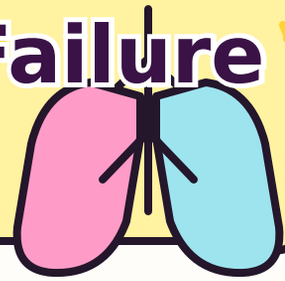
Nursing Care

- Support lung-expanding breathing exercises
- Monitor the patient's breathing, comfort, and response to drainage
- Provide psychosocial support
- Reinforce medication adherence and follow-up plan

Source: Hinkle Chapter 19, slide 45; Chapter 19 antibiotic review

Acute Respiratory Failure





Rapid deterioration in oxygenation, ventilation, or both

Pathophysiology

- May rapidly progress to hypoxemia, hypercapnia, and respiratory acidosis
- Ventilation or perfusion mechanisms are impaired
- The underlying cause must be identified and treated
- Frequent reassessment is essential because status can change quickly

Early Warning Signs

- Restlessness
- Tachycardia or hypertension
- Fatigue and headache
- Accessory-muscle use and decreased breath sounds may be present

Late Danger Signs

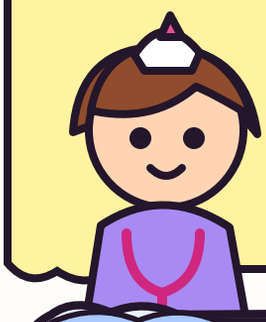
- Confusion or lethargy
- Central cyanosis and diaphoresis
- Respiratory arrest
- Escalate promptly when signs of deterioration appear

Management and Support

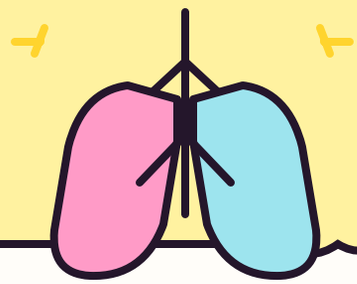
- Treat the underlying cause; intubation and mechanical ventilation may be needed
- Enteral nutrition is preferred when nutritional support is indicated in the slides
- Reduce anxiety and provide a way for the patient to communicate
- Prevent complications with turning, range of motion, mouth care, and skin care

Source: Hinkle Chapter 19, slides 46-47

ARDS



Acute Respiratory Distress Syndrome causes sudden, severe gas-exchange failure



Recognition

- Sudden, progressive pulmonary edema with increasing bilateral lung infiltrates on chest X-ray
- Severe dyspnea and ventilation-perfusion mismatch may develop within 72 hours after a precipitating event
- Hypoxemia is classified by severity and does not respond well to supplemental oxygen alone
- Crackles and intercostal retractions may occur
- The lecture also lists BNP levels as part of the clinical picture

Underlying Mechanism

- The slides describe ARDS without elevated left atrial pressure
- Fluid and inflammation impair oxygen transfer across the lungs
- The patient may deteriorate rapidly after an acute precipitating event
- Treat the cause while supporting oxygenation and circulation

Ventilation and Positioning

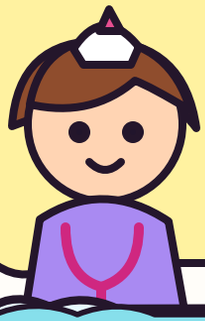
- Intubation and mechanical ventilation with PEEP may help keep alveoli open
- Prone positioning is emphasized for oxygenation
- Reposition frequently to protect skin and other tissues
- Monitor oxygenation and hemodynamic response to the plan

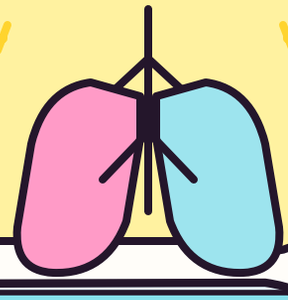
Supportive Care

- Treat the underlying cause and manage hypovolemia to support hemodynamic stability
- Enteral nutrition is preferred when nutritional support is needed
- Anxiety reduction, sedation, or paralysis may be part of the prescribed plan
- Provide ongoing supportive care and prevent immobility-related complications

Source: Hinkle Chapter 19, slides 48-49

Pulmonary Vascular Disorders





The slide deck names these topics but gives limited detail on this slide

Topics Named in the Slides

- Pulmonary edema
- Pulmonary hypertension
- Cor pulmonale
- Pulmonary embolism

Pulmonary Hypertension Note

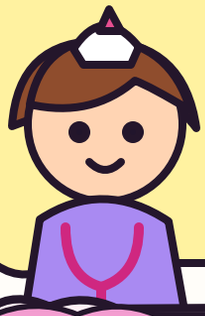
- The slide describes pulmonary arterial pressure above 30 mm Hg with heart failure
- Cor pulmonale is listed as a related topic
- Use your lecture or textbook for the full definition and diagnostic criteria
- Do not use this slide alone to diagnose a patient

Study Focus

- Review the cause and expected effect on oxygenation or right-heart workload from your assigned source
- Connect new dyspnea or oxygenation changes to prompt assessment
- Know the ordered diagnostic and treatment plan for each disorder
- The Chapter 19 slide text supplied here does not provide drug names or a full management plan

Source: Hinkle Chapter 19, slides 50-53

Occupational Lung Disease





Pneumoconioses result from inhaled mineral or inorganic dust

Types and Mechanism

- The slide lists asbestosis, silicosis, and coal workers' pneumoconiosis
- A nonneoplastic lung alteration follows inhalation of mineral or inorganic dust
- Workplace exposure history is central to assessment
- The lecture describes these disorders as preventable, not treatable

Exposure Assessment

- Ask which agent the person encountered
- Determine how long exposure occurred
- Compare the exposure timeline with symptom onset
- Assess whether the symptoms fit the exposure history

Prevention

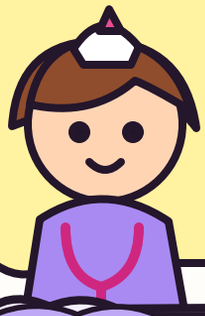
- Reduce or eliminate exposure when possible
- Use protective gear and devices
- Support health education about prevention of lung injury
- Workplace controls and prevention are key themes in the slides

Nursing Role

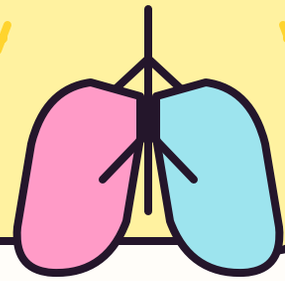
- Serve as an employee advocate
- Provide education about preventive measures
- Document exposure history and related assessment findings
- Support follow-up and symptom management through the care team

Source: Hinkle Chapter 19, slides 54-55

Lung Cancer



Often has no symptoms until later disease stages



Risk and Classification

- The lecture identifies lung cancer as a leading cause of cancer death in the United States
- More than 85 percent is attributed to cigarette smoke in the slide set
- The slides list small-cell and non-small-cell categories
- Slide estimates: 13 percent small-cell and 84 percent non-small-cell tumors
- Stage reflects tumor size, location, lymph-node involvement, and metastasis

Treatment Options in the Slides

- Surgery and radiation
- Chemotherapy and immunotherapy
- Treatment depends on tumor type and stage
- Nurse navigators and palliative care may support the patient through treatment

Symptom Management

- Relieve pain and discomfort and prevent complications
- Manage dyspnea, fatigue, nausea and vomiting, and anorexia
- Support airway clearance and breathing comfort
- Reduce fatigue and address nutrition and function

Whole-Person Nursing Care

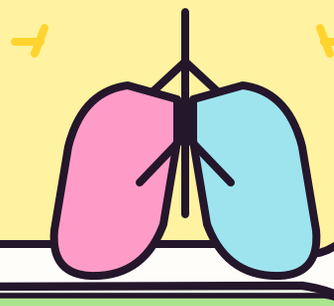
- Provide psychological support
- Assess distress, goals, symptom burden, and caregiver needs
- Coordinate the care team, nurse navigation, and palliative support
- Keep comfort and the patient's priorities visible throughout treatment

Source: Hinkle Chapter 19, slides 56-57

Chest Trauma and Pneumothorax



The slide lists injury patterns and pneumothorax categories



Blunt Trauma

- Sternal or rib fractures
- Flail chest
- Pulmonary contusion
- Assess breathing effort, chest movement, and oxygenation after injury

Penetrating Trauma

- The slide lists penetrating trauma as a major category
- Monitor airway, breathing, circulation, and chest movement
- Escalate acute respiratory distress or unstable status immediately
- Follow emergency and trauma protocols

Pneumothorax Categories

- Simple or spontaneous
- Traumatic
- Tension
- The slide names categories but does not provide full assessment or treatment details

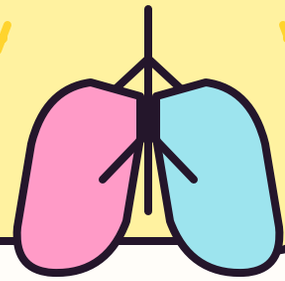
Exam Reminder

- Know the difference between blunt and penetrating trauma
- Know the pneumothorax categories listed by the lecture
- Tension pneumothorax is a time-critical emergency in clinical practice
- Use the assigned lecture or textbook for specific findings and interventions

Source: Hinkle Chapter 19, slides 58-59

Chapter 19 Antibiotic Guide





Drug examples from the supplied NU2106 and Chapter 9 Word documents

Community-Acquired Pneumonia

- Course examples: amoxicillin, amoxicillin-clavulanate, azithromycin, doxycycline, or ceftriaxone
- Clarithromycin is also named in the supplied Chapter 9 review as a macrolide example
- Drug selection depends on the infection, patient risks, allergies, cultures, and prescriber plan
- These are examples from the supplied review, not a universal regimen
- The Chapter 9 review also names clarithromycin; match drug to indication and current guidance

HAP and VAP Examples

- Broader options named: piperacillin-tazobactam, cefepime, vancomycin, or meropenem
- Selection depends on organism and resistance risk
- Review cultures and narrow or change therapy as directed
- Do not treat the list as a fixed combination or automatic bundle

Abscess, Aspiration, and Empyema

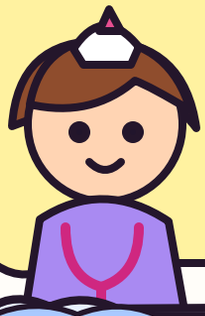
- Ampicillin-sulbactam or amoxicillin-clavulanate are course examples when anaerobes may be involved in lung abscess or aspiration-related bacterial infection
- Empyema therapy is guided by source and cultures; broad IV therapy may start while cultures are pending
- The slide set specifies antibiotics for 4-6 weeks for empyema
- Drainage and treatment of the underlying source are also important

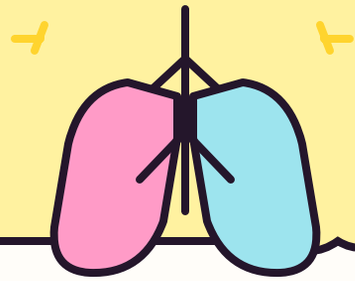
Tuberculosis

- Classic RIPE: rifampin, isoniazid, pyrazinamide, and ethambutol
- The Chapter 9 review also names rifapentine
- TB is not treated with one antibiotic; multidrug adherence is essential
- CDC currently offers shorter regimens for eligible patients; the TB team selects the regimen

Source: NU2106 Exam 4 Chapter 19 Antibiotics.docx; Antibiotics To Review Over For Chapter 9 Antibiotics.docx

Antibiotic Teaching and Safety





Use the right drug for the disease process and the patient

Before and During Therapy

- Check whether the disease process is bacterial and whether antibiotics are indicated
- Check the allergy history and the drug's contraindications
- Obtain ordered cultures before antibiotics when feasible without delaying urgent therapy
- Monitor response, adverse effects, and whether the selected medicine remains appropriate

Viral or Bacterial?

- Most acute pharyngitis is viral; antibiotics are not used for viral illness
- Group A Streptococcus is the most common bacterial cause of pharyngitis
- Doxycycline is listed as an option for adults with penicillin allergy in the review document
- Use testing, clinical guidance, and prescriber orders to decide treatment

Penicillin Allergy Note

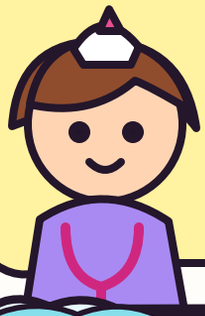
- The course document says a non-IgE-mediated penicillin allergy may allow a first-generation cephalosporin such as cephalexin
- It describes cross-reactivity risk as low
- Clarify the reaction and verify the plan with the prescriber or pharmacist
- Do not infer that a severe immediate allergy is safe to rechallenge

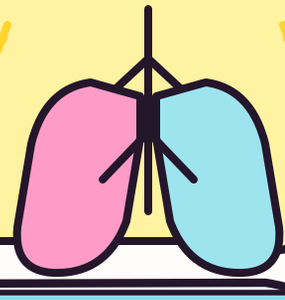
Fluoroquinolone Watch

- Examples named: levofloxacin and moxifloxacin; the review also lists fluoroquinolones as a class
- Serious risks include tendon inflammation or rupture, peripheral neuropathy, central nervous system effects, and QT prolongation
- FDA warns that serious reactions can be disabling; for uncomplicated infections, risks may outweigh benefits
- Teach the patient to promptly report tendon pain/swelling, numbness/tingling, or significant mental-status changes

Source: Antibiotics To Review Over For Chapter 9 Antibiotics.docx; FDA fluoroquinolone label

Quick Compare and Exam Cues





Use these contrasts to organize respiratory assessment and nursing priorities

Atelectasis vs Pneumonia

- Atelectasis: alveoli close or collapse
- Pneumonia: lung parenchyma becomes inflamed from microorganisms
- Both can impair oxygenation and cause decreased respiratory function
- Use assessment, imaging, history, and test results to distinguish them

Pleural Words

- Pleurisy: inflamed pleura with pain linked to breathing
- Pleural effusion: fluid in the pleural space
- Empyema: purulent fluid in the pleural space
- Lung abscess may lead to empyema or a bronchopleural fistula

Respiratory Failure vs ARDS

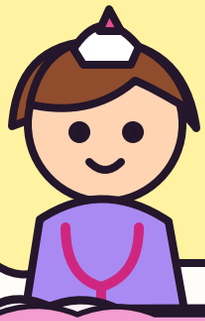
- Respiratory failure: hypoxemia, hypercapnia, or acidosis
- ARDS: acute pulmonary edema and severe hypoxemia that responds poorly to oxygen
- Both need urgent support and treatment of the cause
- Early warning signs include restlessness and increased work of breathing

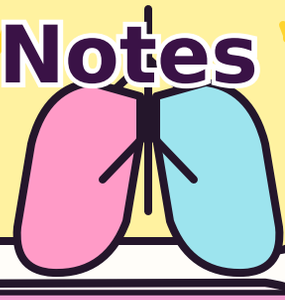
Medication Memory

- Bacterial pneumonia: antibiotic chosen for infection and patient factors; viral pneumonia does not benefit from antibiotics alone
- Lung abscess/aspiration: course examples include ampicillin-sulbactam or amoxicillin-clavulanate when anaerobes may be involved
- TB: classic course cue is multidrug RIPE
- Always connect the medication to indication, contraindications, teaching, and monitoring

Source: Hinkle Chapter 19, slides 3-59; supplied antibiotic reviews

Sources and Scope Notes





Course material stays primary; current safety notes are labeled separately

Course Sources

- Hinkle Chapter 19 lecture: Management of Patients with Chest and Lower Respiratory Tract Disorders, 59 slides
- NU2106 Exam 4 Chapter 19 Antibiotics.docx
- Antibiotics To Review Over For Chapter 9 Antibiotics.docx
- Chapter 20 PowerPoint was excluded from this Chapter 19 guide

Current Official References

- CDC pneumococcal vaccine recommendations, updated February 25, 2026: cdc.gov/pneumococcal/hcp/vaccine-reco
- CDC clinical treatment of TB, April 17, 2025: cdc.gov/tb/hcp/treatment/
- CDC TB adverse events: cdc.gov/tb/hcp/treatment/adverse-even
- FDA fluoroquinolone safety information: accessdata.fda.gov; 2024 ciprofloxacin label

How to Read This Guide

- Named antibiotic regimens are examples taken from the supplied course documents, not treatment orders
- Treatment depends on culture results, resistance, allergies, patient factors, and current local guidance
- Current CDC safety notes are identified where the lecture slide values are older
- Where the slide deck only names a topic, the guide says so instead of inventing missing lecture details

Source: References used to assemble this guide